3 **(a)** Evaluate

$$\sum_{r=1}^{4} \frac{1}{r}$$

giving your answer as a fraction in its lowest terms. **[1]**

(b) Write the sum $1+3+5+7+9$ in a similar way to the series in part **(a)**. **[2]**

(c) Explain why the sum to infinity of $1+3+5+7+9+\ldots$ is **not** well defined. **[1]**